



PROJECT OVERVIEW

To analyse Bright Motors car sales data to uncover key insights that will help the Head of Sales make data-driven decisions to increase sales, optimise inventory and expand the dealership network.

DATA FLOW & ARCHITECTURE

DATA SOURCE



- Bright Motors Car Sales Dataset**
- Vehicle details (Make, Model, Year, Body, Transmission)
 - Sales data (Selling Price, Sale Date)
 - Vehicle condition (Odometer, Condition)
 - Location (State)
 - Seller Information

DATA CLEANING & TRANSFORMATION



- Remove Duplicates (based on VIN)
- Handle missing/null values
- Convert selling price to numeric
- Standardise column names (lowercase, no spaces)
- Convert sale date to date format
- Validate key fields:
 - sellingprice
 - odometer
 - condition

DATA STORAGE (DATABRICKS)



- Structured dataset stored in Databricks
- Cleaned and transformed data
- Ready for SQL querying
- Single source of truth for analysis

EXPLORATORY DATA ANALYSIS (SQL)



- Sales by:**
 - Make & Model
 - State (Region)
- Price Analysis:**
 - Price vs Odometer
 - Price vs Year
- Sales Trends:**
 - Overtime (monthly/yearly)
- Inventory Insights:**
 - Body type distribution
 - Transmission trends
- Seller Performance**

DATA VISUALISATION & REPORTING



- Pivot Tables**
- Revenue by Make/Model
 - Sales by State
 - Sales by Year/Month
- Charts**
- Line Chart - Sales trends
 - Bar Chart - Top car brands
 - Scatter plot - Price vs Mileage
 - Donut Chart - Body type distribution
- Slicers/Filters**
- State
 - Year
 - Body Type
 - Transmission

BUSINESS INSIGHTS PRESENTATION



- Key Findings**
- Top Performing Vehicles (Make/Model)
 - High-performing regions (State)
 - Customer buying patterns & trends
 - Pricing & mileage insights
- Strategic Recommendations**
- Inventory optimisation
 - Pricing strategy
 - Regional expansion
 - Sales and marketing focus

FEATURE ENGINEERING



- Extract**
- Sale Month
 - Sale Year



- Create Analysis Categories**
- Price Ranges
 - Mileage Ranges



- Create Proxy Metric**
- Estimated Profit = Selling Price - MMR (if used)

Additional fields for deeper analysis

Raw, unprocessed transactional dataset

Cleaned, standardised and enriched data ready for storage

Reliable and consistent data foundation

Aggregations, grouping and trend analysis

Interactive dashboards for deep insights

Presented to Head of Sales for decision-making

KEY OUTCOME

Actionable insights for smarter business decisions

CONCLUSION

data-driven insights empower Bright Motors to optimise inventory, improve pricing strategies and expand dealership network for higher profitability and sustainable growth.



Source



ETL Process



Storage



SQL Analysis



Visualisation



Presentation